Gradient Lab | Simulation - Linear Equations and Gradients

Overview

Gradient Lab is a single-page, browser-based simulation with three independent levels (Beginner, Intermediate and Advanced) that a student can enter in any order. It supports lesson MTH12_09_02_01, Linear Equations and Gradients.

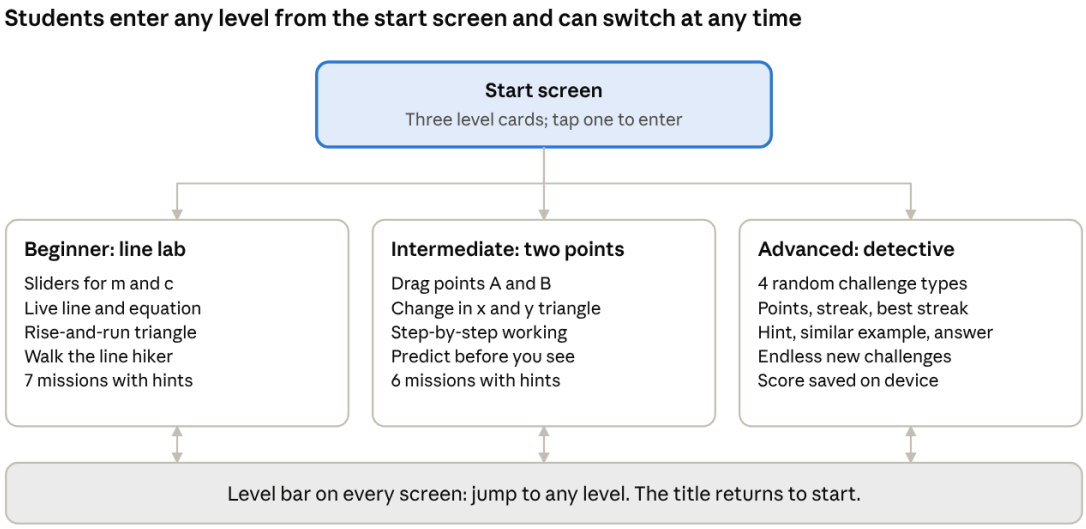
Item	Detail
Audience	Students who have watched the lesson on y = mx + c and the gradient formula
Entry points	Start screen with three level cards, plus a level bar on every screen
Total missions	13 fixed missions (Beginner 7, Intermediate 6), plus unlimited Advanced challenges
Grid	Coordinates from −8 to 8 on both axes, whole-number snapping
Platform	One HTML file; runs in any modern browser on phone, tablet or desktop
Themes	Light and dark, following the device setting
Progress	Saved on the student's device (no login)

By the end, students can:

1. Say whether a line has a positive, negative or zero gradient from its direction.
2. Explain what m and c do in y = mx + c.
3. Calculate a gradient from two points, including negatives and fractions.
4. Show that the gradient is the same for any two points on a straight line.
5. Find the equation of a line from a graph or from two points, and find missing coordinates.

Screen map and navigation

There are four screens: the start screen and one screen per level. The level bar at the top lets a student move between any two levels in one tap.



screen map · start screen, 3 levels, level bar

On first visit the start screen shows. On a return visit the simulation opens on the last level used. Each level keeps its own state while the student switches between levels in the same session.

Shared components

All three levels are built from the same six parts, so they look and behave the same way.

Component	What it is	Behaviour
Level bar	Fixed bar at the top: the title “Gradient lab” and three level tabs	The current level is highlighted. Beginner and Intermediate tabs show missions done (e.g. 3/7); Advanced shows challenges solved. The title returns to the start screen.
Coordinate grid	Square grid from −8 to 8 on both axes, numbered every 2 units	Red line for the student’s line, dashed grey for a reference line. On wide screens it stays in view while the panel scrolls; on phones it sits above the panel.
Level panel	Right-hand column with the level’s title, controls, working and missions	Updates live as the student moves anything.
Mission bar	Numbered square buttons, one per mission	Current mission outlined, completed missions green. Tapping a number jumps to that

Component	What it is	Behaviour
		mission, so missions can be done in any order.
Mission card	Mission text, “Give me a hint” button, hint note	Missions are checked only after the student moves something, never on page load. On success: “Mission complete!” note and a “Next mission” button. After the last mission: “You finished every mission at this level. Try the next level.”
Feedback notes	Coloured boxes with a handwritten-style heading	Yellow = hint, blue = similar example, green = correct, red = answer or mistake.

Answer input rules

Answer boxes accept whole numbers, decimals, negatives and fractions. Both “-2” and “−2” work, and “3/2” and “1.5” are treated as the same value. Pressing Enter is the same as pressing Check. If the box is empty or not a number, a yellow note explains what to type and no attempt is counted.

Number display rules

Fractions are shown stacked (numerator over denominator). Negative numbers inside working are put in brackets, for example 3 − (−1). Equations are written in their simplest form: y = x + 2, not y = 1x + 2; y = −3x, not y = −3x + 0.

Beginner level: the line lab

The student controls the line with two sliders and sees y = mx + c, the graph and a plain-English explanation change together.

Layout

Left: the grid, with a caption line under it for the hiker. Right: the panel, from top to bottom: title, live equation, two sliders, three controls, explanation panel, missions.

Controls

Control	Type	Range or default	Effect
m, the gradient	Slider	−4 to 4, steps of 0.5; starts at 0	Changes the slope of the red line and the equation

Control	Type	Range or default	Effect
c, the y-intercept	Slider	−6 to 6, steps of 1; starts at 0	Moves the line up or down; a dot marks (0, c)
Walk the line	Button	—	Sends the hiker along the line
Show rise and run	Checkbox	On	Shows the shaded triangle
Show the lesson's line	Checkbox	Off	Adds y = 2x − 4 as a dashed grey line

Behaviours

- **Live equation:** shown in large type above the sliders, simplified (y = 0, y = x, y = ½x + 1, y = −3x + 2).
- **Rise-and-run triangle:** starts at (0, c). A blue line goes across 1 square, labelled "across 1", and a green line goes up or down m squares, labelled "up m" or "down m". When m is a half value, the run is 2 so the rise is a whole number. Hidden when m = 0.
- **Walk the line:** a blue dot travels from x = −8 to x = 8 in about 3.2 seconds. It fades while it is off the grid. The caption reads "Climbing uphill…", "Walking downhill…" or "Walking on flat ground…", then ends with "That was uphill: m is positive.", "That was downhill: m is negative." or "Flat all the way: m is 0." With reduced motion on, the dot jumps straight to the end.
- **Explanation panel:** three sentences that update live. Sentence 1 states the sign of m, the direction and the steepness, e.g. "m = 2 is positive, so the line climbs uphill from left to right. It is steep." Sentence 2 gives the rise and run, e.g. "For every 1 square across, the line goes up 2." Sentence 3 gives the intercept, e.g. "c = −4, so the line crosses the y-axis at (0, −4)."

Size of m	Steepness word
Less than 1.5	gentle
1.5 to less than 3	steep
3 or more	very steep

Missions

A mission is checked each time a slider moves.

No.	Mission	Completed when	Hint	Success message
1	Make the line climb uphill from left to right.	m > 0	Uphill lines have a positive gradient.	A positive gradient makes

No.	Mission	Completed when	Hint	Success message
			Move the m slider to the right of 0.	the line rise from left to right.
2	Make the line go downhill from left to right.	$m < 0$	Downhill lines have a negative gradient. Move m below 0.	A negative gradient makes the line fall from left to right.
3	Make the line completely flat.	$m = 0$	A flat line neither climbs nor falls. What value of m would that be?	When m = 0, the line is flat. It does not go up or down.
4	Make the line cross the y-axis at 3.	$c = 3$	c is the y-intercept. It tells you where the line crosses the y-axis.	c = 3, so the line crosses the y-axis at (0, 3).
5	Build the line from the lesson: $y = 2x - 4$.	$m = 2$ and $c = -4$	Match the equation to $y = mx + c$. What is m? What is c?	m = 2 and c = −4. This is the line through A (5, 6) and B (3, 2) from the lesson.
6	Make a gentle uphill line, $y = ½x + 1$.	$m = 0.5$ and $c = 1$	A gradient of ½ means the line goes up 1 for every 2 steps across.	With m = ½, the line rises 1 square for every 2 squares across. It is less steep than m = 1.
7	Make a downhill line steeper than $y = -2x + 1$.	$m < -2$	Steeper downhill means a bigger negative number. Try something further from 0 than −2.	The further m is from 0, the steeper the line. m = −3 is steeper than m = −2.

Intermediate level: two-point gradient

The student drags two points, A and B, and the line, triangle and working update as each point moves. There are no sliders on this level.

Layout

Left: the grid with points A (blue) and B (purple) and a caption line. Right: the panel, from top to bottom: title, two checkboxes, working panel, missions.

Controls

Control	Type	Range or default	Effect
Point A	Draggable point	Starts at (−3, −1); x and y from −7 to 7	Moves one end of the line
Point B	Draggable point	Starts at (1, 3); x and y from −7 to 7	Moves the other end of the line
Predict before you see	Checkbox	Off	Hides the working until the student predicts
Show the equation	Checkbox	Off	Adds the c calculation and the full equation

Points snap to whole-number coordinates and show their label and coordinates, e.g. A (−3, −1). A selected point can also be moved one square at a time with the arrow keys. A point cannot be dropped on top of the other point.

Behaviours

- **Line:** the red line passes through A and B. If A and B have the same x-coordinate, a vertical line is drawn.
- **Triangle:** shaded yellow, from A across to below or above B, then up or down to B. The blue side is labelled “across __” and the green side “up __” or “down __”. Labels sit outside the triangle and move to the other side near the grid edge.
- **Working panel:** shows the two points, then the gradient line by line. The y-values are green and the x-values are blue, to match the triangle.

```
\text{Gradient} = \frac{\text{change in } y}{\text{change in } x} = \frac{3 - (-1)}{1 - (-3)} = \frac{4}{4} = 1
```

The fraction is simplified and the sign moved to the top when needed. A sentence below says whether the line climbs uphill, goes downhill or is flat. With “Show the equation” on, the panel adds “Using A: c = y − mx = …” and the full equation in large type.

- **Vertical line:** the working ends with a denominator of 0, and a red note says: “Division by zero. The change in x is 0, so the line is vertical. You cannot divide by 0, so this line has no number for its gradient. Move A or B sideways.”

Predict mode

With “Predict before you see” on, the working is replaced by the question “The line passes through A (x, y) and B (x, y). What is its gradient?”, filled with the real coordinates. Below it are an answer box, Check and “Just show me”. After any answer, the working appears below the feedback. Moving a point hides the working again for a new prediction.

Student's answer	Feedback heading	Feedback text (values filled in)
Correct	Spot on!	You predicted [answer], and the gradient is [gradient].
Change in x ÷ change in y	Upside down	You found change in x ÷ change in y. The gradient is the other way up: [gradient].
Right size, wrong sign	Check the sign	The size is right but the sign is not. This line rises/falls, so the gradient is [gradient].
Any other wrong answer	Not quite	You predicted [answer]. The working below shows how to get [gradient].
Vertical line	Careful	The change in x is 0, so you cannot divide by it. This vertical line has no number for its gradient.

Missions

A mission is checked each time a point is dropped or moved with an arrow key.

No.	Mission	Completed when	Hint	Success message
1	Drag A and B so the gradient is 2.	gradient = 2	A gradient of 2 means the change in y is twice the change in x. Try moving 1 across and 2 up.	Change in y ÷ change in x = 2. The line goes up 2 for every 1 across.
2	Make the gradient −1.	gradient = −1	The line must fall from left to right, going down 1 for every 1 across.	The line falls 1 for every 1 across, so the gradient is −1.
3	Make the gradient ½.	gradient = ½	Change in y must be half of the change in x. Try 2 across and 1 up.	Up 1 for every 2 across gives a gradient of ½.
4	Same line, different points. A and B now stay on	3 different pairs tried	Drag A or B along the dashed line. Each new position	Every pair of points gave a gradient of 2. The

No.	Mission	Completed when	Hint	Success message
	y = 2x − 4. Try 3 different pairs of points and watch the gradient.		makes a new pair of points.	gradient of a straight line is the same for any two points on the line.
5	Make a line with gradient −3 that crosses the y-axis at 2.	gradient = −3 and c = 2	Put one point at (0, 2). Then move the other 1 across and 3 down.	Gradient −3 and y-intercept 2: this is the line y = −3x + 2.
6	Make the gradient ⅔ using points that are not on the y-axis.	gradient = ⅔, and neither point has x = 0	Change in y ÷ change in x = 2 ÷ 3. Try 3 across and 2 up, away from the y-axis.	Up 2 for every 3 across gives a gradient of ⅔.

Mission 4 changes how the points move. While it is open, y = 2x − 4 is drawn as a dashed line and both points are locked onto it, with x from −1 to 5. A counter shows “Pairs tried: 1 of 3”, rising to 3. A pair is counted when its two x-values have not been used together before. The caption changes to “A and B can only move along y = 2x − 4.”

Advanced level: equation detective

The Advanced level generates a new random challenge each time, in one of four types. Every challenge gives three tries, with more help after each wrong answer.

Layout

Left: the grid and a caption line. Right: the panel, from top to bottom: title, challenge type buttons, four score boxes (points, solved, streak, best streak), the challenge, then “New challenge” and “Reset my score”.

The type buttons are Mixed (the default), Read the graph, Two points, Missing coordinate and Hit the targets. Mixed picks one of the four types at random for each challenge. Choosing a type loads a fresh challenge straight away.

Challenge types

Type	Question shown	Graph	Answer	Generation rules
Read the graph	Find the equation of the red line. Give m and c.	Red line, no points marked	Two boxes: m and c	Gradient from ±1, ±2, ±3, ±½, ±3/2, ±⅓, ±⅔. c is a whole number from −4 to 4. The line passes exactly through a second

Type	Question shown	Graph	Answer	Generation rules
				grid point that is on screen.
Two points	A straight line passes through (x_1, y_1) and (x_2, y_2). Find its equation in the form y = mx + c.	None while working; caption "No graph this time. Use the coordinates." The line and points appear after the challenge ends, if they fit.	Two boxes: m and c	Same gradients as above. c from −5 to 5. Both points have whole-number coordinates.
Missing coordinate	The point (k, y) lies on the line y = mx + c. Find k. Or: the point (x, y) lies on the line. Find y.	The line	One box: k or y	Gradient a whole number from ±1 to ±4. c from −6 to 6. k from −5 to 5, not 0. Each version appears half the time.
Hit the targets	Set m and c so the line passes through both target dots, then press Check.	Two ring targets with coordinates; a ring turns green when the line passes through it	Sliders: m from −4 to 4 in steps of 0.5; c from −8 to 8 in steps of 1	Gradient from ±0.5 to ±3 in steps of 0.5. c from −4 to 4. Targets at x = −4, −2, 2 or 4, both on screen.

Hints

Type	Hint
Read the graph	c is where the line crosses the y-axis. For m, find two places where the line passes exactly through grid corners, then work out change in y ÷ change in x.
Two points	First find the gradient: change in y ÷ change in x. Then put one point into y = mx + c and solve for c.
Missing coordinate	Replace y and x with the numbers you know, then solve; or put the x-value in, multiply first, then add or subtract c.

Type	Hint
Hit the targets	Work out the gradient between the two dots first and set m. Then move c until the line passes through one dot. It will pass through both.

Feedback ladder

1. **Correct, on any try:** green “Correct!” note with the full working and the points earned. The answer locks and a “Next challenge” button appears.
2. **First wrong answer:** yellow “Not yet” note, a specific comment where one applies, then the hint and “Try again.”
3. **Second wrong answer:** the hint stays, and a blue “Try a similar one first” note shows a new challenge of the same type with different numbers, fully worked. Read the graph and Hit the targets examples include a small graph. It ends with “Now use the same steps and try again.”
4. **Third wrong answer:** red “Here is the answer” note with the full working. For Hit the targets, the sliders move to the correct values. The answer locks and “Next challenge” appears.

The specific comments on a first wrong answer are: “Your gradient is right, but c is not.”, “Your c is right, but the gradient is not.”, “Check the sign of your gradient.”, and for targets, “Your line passes through 1 of the 2 dots.”

Scoring

Result	Points	Streak
Correct on the 1st try	+3	Goes up by 1
Correct on the 2nd try	+2	Resets to 0
Correct on the 3rd try	+1	Resets to 0
Answer revealed	0	Resets to 0

Best streak keeps the highest streak reached. “New challenge” skips without changing the score or streak. “Reset my score” sets all four boxes back to 0.

Sample working (Two points)

For the points (4, −4) and (−4, 8):

```
\text{Gradient} = \frac{8 - (-4)}{-4 - 4} = \frac{12}{-8} = -\frac{3}{2}
```

Substitute (4, −4): −4 = −3/2 × 4 + c, so c = −4 − (−6) = 2. The equation is y = −3/2 x + 2.

Saving progress and accessibility

Progress is saved in the browser on the student’s own device, with no login. A different device or browser starts fresh, and teachers cannot see students’ results.

Saved item	Used for
Last level opened	Opening on that level on the next visit
Completed Beginner missions	Green mission buttons and the count on the Beginner tab
Completed Intermediate missions	Green mission buttons and the count on the Intermediate tab
Advanced points, solved, streak and best streak	Score boxes and the count on the Advanced tab
Advanced challenge type	Reopening on the same type

If the browser blocks saving, the simulation still works but starts fresh on each visit. Slider positions, point positions and the current challenge are not saved.

Accessibility

- Points A and B can be moved with the arrow keys as well as by dragging, and each announces its position to screen readers.
- Every button, slider and answer box has a visible focus outline and a label.
- Feedback and the live equation are announced to screen readers when they change.
- Colour is never the only signal: green and red notes also carry a heading, such as “Correct!” or “Here is the answer”.
- The hiker animation respects the device’s reduced-motion setting.
- Light and dark themes follow the device setting.
- The layout changes to one column on screens narrower than 900 pixels.